Jie Shen

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Education

University of Toronto — *September 2022 – April 2027*
Honours B.Sc. in Mathematics and Statistics, Minor in Computer Science — Mississauga, ON
Relevant Coursework: Machine Learning, Statistical Learning, Regression Analysis, Data Structures & Algorithms, Software Design, Probability, Statistical Inference

Experience

Project Anomaly - Independent Apparel Brand — *May 2026 – Present*
Full-Stack Developer — Toronto, ON

- Developing the official e-commerce platform for an independent apparel brand, translating business requirements and visual concepts into a responsive web application.
- Designing the frontend, backend API, and database architecture to support product catalogues, shopping carts, inventory, customer orders, and Stripe-based checkout workflows.
- Collaborating with the founder on technical requirements, interface design, feature prioritization, and launch planning.

Projects

AI Study Assistant — *Summer 2026*
React, Vite, JavaScript, Python, FastAPI, FAISS, SentenceTransformers, PyMuPDF, Ollama

- Built a **full-stack Retrieval-Augmented Generation (RAG)** application that indexes uploaded PDFs and answers natural-language questions using semantically retrieved document context.
- Engineered an end-to-end retrieval pipeline with **PyMuPDF extraction**, overlapping text chunking, SentenceTransformers embeddings, cosine-similarity search, and **FAISS vector indexing**.
- Developed a **React/Vite frontend** with PDF upload, document-ready state, provider selection, loading/error handling, and asynchronous integration with a **FastAPI REST backend**.
- Designed a modular LLM provider abstraction supporting **local Ollama/Llama 3.2 inference** and optional OpenAI integration without modifying the underlying retrieval pipeline.

Interactive Canvas Studio — *Summer 2025; Refactored in 2026*
Java, JavaFX, MVC-Style Architecture, Factory Pattern, Object-Oriented Design

- Developed a JavaFX desktop drawing application with **8 drawing tools**, real-time mouse-driven rendering, colour and fill controls, adjustable line thickness, and light/dark display modes.
- Designed an extensible **MVC-style architecture** that separates application state, rendering, and user-input handling into modular model, view, and controller components.
- Implemented a polymorphic shape hierarchy with factory-based creation for circles, rectangles, squares, ovals, triangles, polylines, freehand sketches, and wave tools.
- Built undo/redo, cut/copy/paste, and custom `.canvas` save/load persistence while maintaining modular components and architecture documentation for future feature expansion.

Hybrid NLP & Behavioural Classification Engine — *Winter 2026*
Python, NumPy, Pandas, NLP, Multinomial Logistic Regression, Machine Learning

- Built an end-to-end **3-class classification system** combining Bag-of-Words text representations with normalized behavioural survey variables.
- Implemented multinomial logistic regression **from scratch** using vectorized gradient descent, softmax activation, cross-entropy loss, and L2 regularization.
- Achieved approximately **88% multiclass classification accuracy** through hybrid feature engineering, hyperparameter tuning, and comparison with Naive Bayes and Random Forest models used as comparative baselines.
- Structured the system into reusable preprocessing, feature engineering, training, inference, tuning, and evaluation modules to support reproducible experimentation.

Technical Skills

Languages: Python, Java, JavaScript, SQL, R, HTML, CSS
Frameworks & Tools: FastAPI, JavaFX, scikit-learn, NumPy, Pandas, FAISS, Maven, JUnit, Git, GitHub, LaTeX
Core Skills: Object-Oriented Programming, MVC, REST APIs, Event-Driven Programming, Data Structures & Algorithms, Machine Learning, NLP, Classification, Regression, Cross-Validation, Feature Engineering